Job Analytics Portal Report

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Introduction

This report documents three advanced data visualization tasks focused on analyzing job-related data using conditional filtering, time-based rendering, and stylistic customization for dashboard deployment.

Background

With the increasing need for interactive dashboards, data visualization plays a vital role in presenting filtered, meaningful insights. These tasks were designed to enhance understanding of data relationships and improve dynamic dashboard interactions based on complex business rules.

Learning Objectives

- Understand multi-dimensional data relationships (e.g., between countries, job titles, and roles).
- Learn to apply complex filters and conditions on visualizations.
- Implement time-based conditional rendering for charts.
- Enhance front-end visualization with custom styling and user-based preferences.

Activities and Tasks

Task 1:
Create a chart showing the relationship between country, job title, and role.
Objective: Understand hierarchical relations and aggregation.

Task 2:
Draw a chart for preference vs work type, filtered by multiple strict conditions:
- Work type = ‘Intern’, latitude < 10, country name not starting with A–D,
- Job title ≤ 10 characters, company size < 50,000,
- Only show the chart between 3PM to 5PM IST.

Task 3:
Draw a conditional chart showing job roles in India (orange) and Germany (green):
- Filtered by:
- Qualification = B.Tech, Work Type = Full-time
- Experience > 2 years, Job Title in Data Scientist, Art Teacher, AeroSpace Engineer
- Salary > $10k, Job Portal = Indeed, Preference = Female
- Posting Date < 08/01/2023, visible only 3PM–5PM IST

Skills and Competencies Developed

- Complex filter logic and data preprocessing.
- Use of conditional rendering in dashboard tools (e.g., using JavaScript or backend logic to hide/show).
- Styling data visualizations with color rules.
- Time-based UI control using date/time libraries.
- Effective communication through visual mediums.

Feedback and Evidence

- Charts successfully responded to real-time clock filtering (time-controlled visibility).
- Applied filters yielded correct datasets and expected visual outputs.
- Stakeholders confirmed visual clarity and relevance of conditions.
- Code snippets and screenshots of final dashboards were saved for portfolio use.

Challenges and Solutions

Challenges

- Implementing time-based chart visibility.
- Applying multiple complex filters.
- Color coding specific countries.
- Handling large datasets efficiently.

Solutions

- Used JavaScript (e.g., with Date object) to dynamically render/hide charts.
- Broke conditions into smaller segments for testing and validation.
- Used conditional color logic in the chart configuration.
- Applied backend filtering and sent only required data to the frontend.

Outcomes and Impact

- Developed expertise in building intelligent, real-world dashboards.
- Understood importance of audience-specific data filtering.
- Practiced combining design, logic, and performance for professional reporting dashboards.
- Increased confidence in integrating business rules with UI/UX requirements.

Conclusion

These tasks provided in-depth exposure to conditional data visualization, a critical skill in web development and data analytics. With successful implementation of filters, conditional time display, and styling, the outcomes align with real-world expectations for job-market intelligence dashboards.